1) Assuming no loss, applying Bernoulli's equation.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

where $P_1 = P_2 + \rho g h$

$$Q = A_1 V_1 = A_2 V_2$$

$$V_1 = \frac{4Q}{\pi D^2} \qquad V_2 = \frac{4Q}{\pi d^2}$$

$$z_2 - z_1 = L \sin\theta$$

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\frac{\cancel{P_2} + \rho g h}{\cancel{\rho g}} + \frac{\left(\frac{4Q}{\pi D^2}\right)^2}{2g} + z_1 - \frac{\cancel{P}}{\cancel{\rho g}} - \frac{\left(\frac{4Q}{\pi d^2}\right)^2}{2g} - z_2 = 0$$

$$h + \frac{16Q^2}{2g\pi^2 D^4} + \underbrace{\cancel{z_1 - z_2}}_{-L\sin\theta} - \frac{16Q^2}{2g\pi^2 d^4} = 0$$

$$\frac{16Q^2}{2g\pi^2}\left(\frac{1}{D^4} - \frac{1}{d^4}\right) + h - L\sin\theta = 0$$

1) $$\frac{16Q^2}{2g\pi^2}\left(\frac{1}{D^4}-\frac{1}{d^4}\right)+h-L\sin\theta=0$$

$$\frac{16Q^2}{2g\pi^2}\left(\frac{1}{d^4}-\frac{1}{D^4}\right)=h-L\sin\theta$$

$$\frac{16Q^2}{\pi^2 d^4}\left(\frac{d^4}{d^4}-\frac{d^4}{D^4}\right)=2g(h-L\sin\theta)$$

$$\frac{16Q^2}{\pi^2 d^4}\left(1-\beta^4\right)=2g(h-L\sin\theta) \quad \because \beta=\frac{d}{D}$$

$$\frac{16Q^2}{\pi^2 d^4}=\frac{2g(h-L\sin\theta)}{1-\beta^4}$$

$$\frac{4Q}{\pi d^2}=\sqrt{\frac{2g(h-L\sin\theta)}{1-\beta^4}}$$

$$Q=\frac{\pi d^2}{4}\sqrt{\frac{2g(h-L\sin\theta)}{1-\beta^4}}$$

The above is the ideal flow rate in the pipe.

The actual flow in the pipe, with loss, is:

$$Q=\frac{C_v\pi d^2}{4}\sqrt{\frac{2g(h-L\sin\theta)}{1-\beta^4}} \quad \text{(shown)}$$

2) Assuming no losses, using the Bernoulli equation:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$Q = A_1 V_1 = A_2 V_2$$

$$V_1 = \frac{Q}{A_1} \quad , \quad V_2 = \frac{Q}{A_2}$$

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\frac{P_1}{\rho g} + \frac{Q^2}{2g A_1^2} + z_1 = \frac{P_2}{\rho g} + \frac{Q^2}{2g A_2^2} + z_2$$

$$\frac{Q^2}{2\cancel{g}}\left(\frac{1}{A_1^2} - \frac{1}{A_2^2}\right) = \frac{P_2 - P_1}{\rho \cancel{g}} + z_2 - z_1$$

$$\frac{Q^2}{A_2^2}\left(1 - \beta^4\right) = \frac{2(P_2 - P_1)}{\rho} + z_2 - z_1$$

$$\frac{Q}{A_2} = \sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + z_2 - z_1}{1 - \beta^4}}$$

$$Q = A_2\sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + z_2 - z_1}{1 - \beta^4}}$$

2) $\therefore$ The ideal flow rate is:

$$Q_{ideal} = A_2 \sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + z_2 - z_1}{1 - \beta^4}}$$

The actual flow rate:

$$Q = C_0 A_2 \sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + z_2 - z_1}{1 - \beta^4}}$$

For a horizontal pipe:

$$Q = C_0 A_2 \sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + \cancelto{0}{z_2 - z_1}}{1 - \beta^4}}$$

$$= C_0 A_2 \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - \beta^4)}}$$

$$= \frac{C_0 \pi d^2}{4} \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - \beta^4)}}$$

2) $Re = \dfrac{V_1 D}{\nu} = \dfrac{\left(\frac{4(0.03)}{\pi(0.08)^2}\right)\left(8\times10^{-2}\right)}{1.12\times10^{-6}}$

$= 4.26307883\times10^5$

$\beta = \dfrac{d}{D} = \dfrac{5}{8}$

$= 0.625$

From Figure S1:

$C_o \approx 0.608$

$\therefore Q = \dfrac{C_o \pi d^2}{4}\sqrt{\dfrac{2(P_1-P_2)}{\rho(1-\beta^4)}}$

$0.03 = \dfrac{0.608\pi(5\times10^{-2})^2}{4}\sqrt{\dfrac{2(P_1-P_2)}{1000\left(1-\left(\frac{5}{8}\right)^4\right)}}$

$25.12972786 = \sqrt{\dfrac{2(P_1-P_2)}{847.4121094}}$

$P_1-P_2 = 267.5717388\times10^3 Pa$

$\approx 268\ kPa$

3) From question 2:

$$Q_{ideal} = A_2 \sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + z_2 - z_1}{1 - \beta^4}}$$

$$Q = \frac{C_n \pi d^2}{4} \sqrt{\frac{\frac{2(P_1 - P_2)}{\rho} + z_2 - z_1}{1 - \beta^4}}$$

Since the pipe is horizontal:

$$Q = \frac{C_n \pi d^2}{4} \sqrt{\frac{2(P_1 - P_2)}{\rho(1-\beta^4)}}$$

$$P_1 - P_2 = \rho g h$$

$$= 1000 \times 9.81 \times 7.3 \times 10^{-3}$$

$$= 71.613 \text{ Pa}$$

$$\beta = \frac{d}{D} = \frac{50}{80} = 0.625$$

$$Q = \frac{C_n \pi (50 \times 10^{-3})^2}{4} \sqrt{\frac{2(71.613)}{1.23(1 - (\frac{5}{8})^4)}}$$

$$Q = 0.02301658743$$

$$\approx 0.023 C_n$$

3) $Re = \frac{V_1 D}{\nu} = \frac{\frac{0.023 C_n}{\frac{\pi (80\times10^{-3})^{\cancel{2}}}{4}} \cancel{(80\times10^{-3})}}{1.46\times10^{-5}}$

$= 2.509043604\times10^{4} C_n$

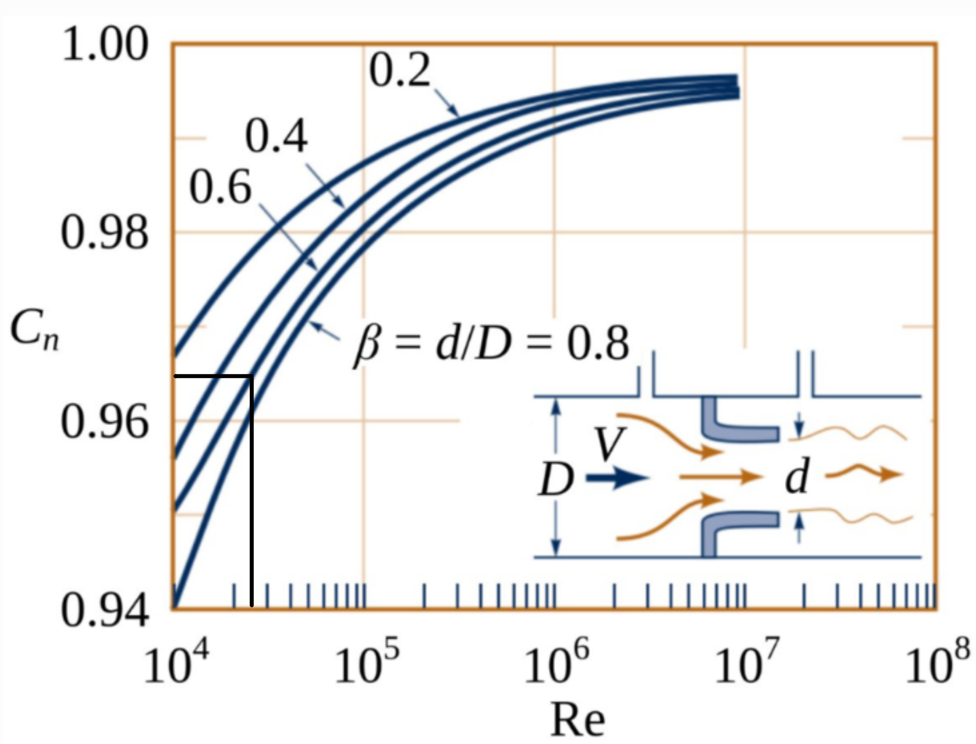
$\approx 2.51\times10^{4} C_n$

Try $C_n = 1$,

$Re = 2.51\times10^{4}(1)$

$= 2.51\times10^{4}$

Reading C_n from the graph,

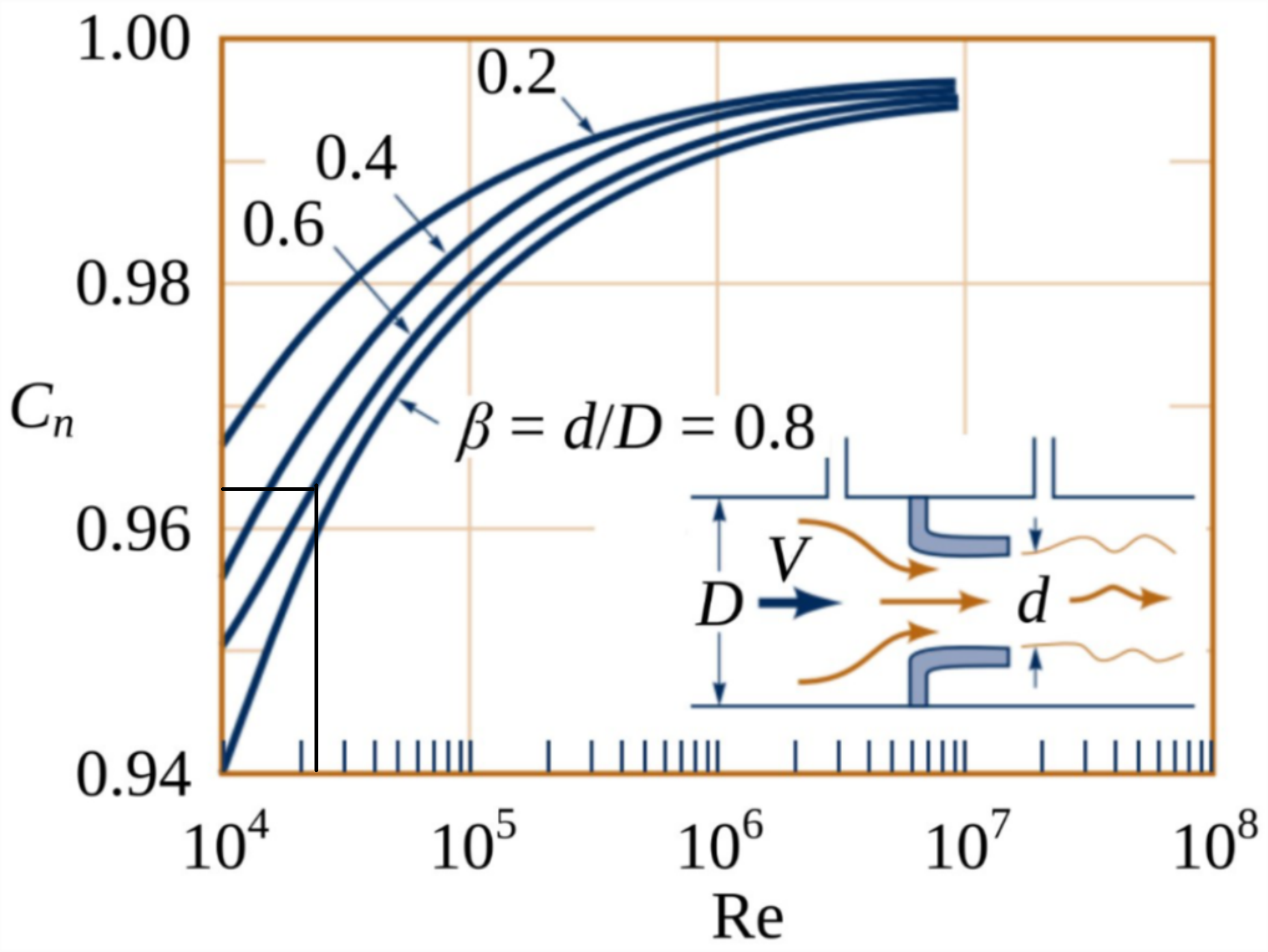


$C_n \approx 0.965$

3) Trying $C_n = 0.965$

$$Re = 2.51 \times 10^4 \times 0.965$$
$$= 2.421227078 \times 10^4$$

Reading C_n from the graph,

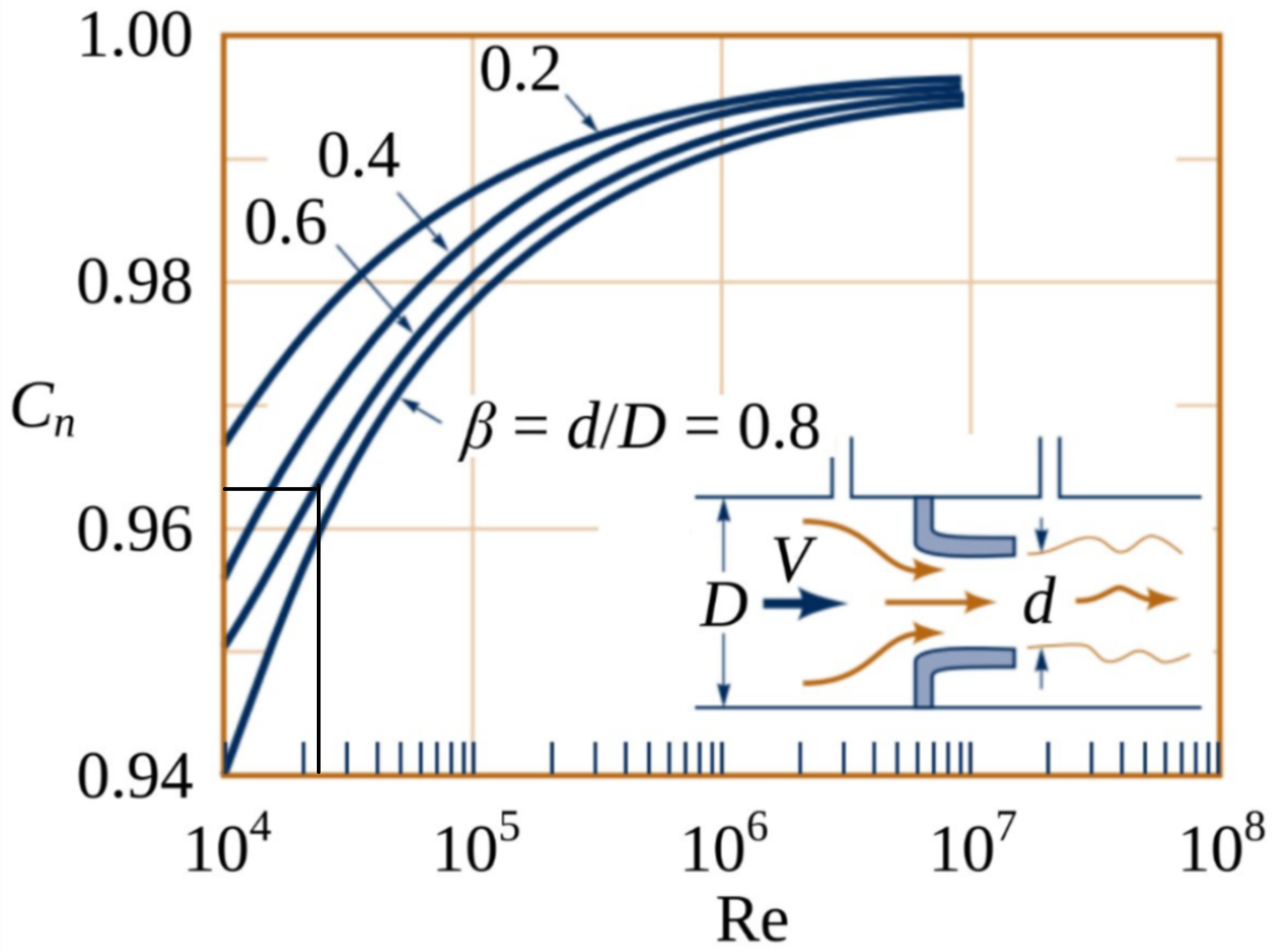


$C_n = 0.963$

Trying $C_n = 0.963$,

$$Re = 2.51 \times 0.963$$
$$= 2.41620899 1 \times 10^4$$

3) Reading C_n from the graph,



$C_n = 0.963$

Since $C_n = 0.963$ is the same as the previous C_n,

$C_n = 0.963$,

$\therefore Q = 0.023 \times 0.963$

$= 0.0221649737$

$\approx 0.0222 \, m^3 s^{-1}$